Xiaohan Zhang

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RESEARCH SUMMARY

Robotics and Artificial Intelligence

My goal is to develop learning and planning algorithms that enable autonomous robots to perform service tasks in home environments. In particular, I'm interested in: 1) Learning for integrated task and motion planning. 2) Foundation models for long-horizon planning. 3) Coordinated whole-body motion generation. 4) Multimodal and interactive robot perception.

ACADEMIC BACKGROUND

State University of New York at Binghamton, Binghamton, NY
Ph.D. in Computer Science — Aug. 2020 – Present
- Advisor: Shiqi Zhang

State University of New York at Binghamton, Binghamton, NY
M.S. in Computer Science — Aug. 2019 – May 2020

Renmin University of China, Beijing, China
B.E. in Computer Science — Sep. 2015 – May 2019

EMPLOYMENT

Embodied AI, FAIR, Meta, Pittsburgh, PA
Research Intern — May. 2023 – Present
- Advisor: Chris Paxton

Learning Agents Research Group, UT Austin, Austin, TX
Visiting Researcher — Jan. 2023 – Apr. 2023
- Advisor: Peter Stone

Robotics, Google Brain, Google, Mountain View, CA
Student Researcher — Mar. 2022 - Aug. 2022
- Advisors: Montserrat Gonzalez and Fei Xia

Autonomous Intelligent Robotics Lab, SUNY Binghamton, Binghamton, NY
Research Assistant — Dec. 2020 - Present

Department of Computer Science, SUNY Binghamton, Binghamton, NY
Teaching Assistant — Jan. 2020 - Dec. 2020

PREPRINTS

1. Priyam Parashar, Vidhi Jain, **Xiaohan Zhang**, Jay Vakil, Sam Powers, Yonatan Bisk, and Chris Paxton. SLAP: Spatial-Language Attention Policies. *ArXiv, 2023*
2. Bo Liu*, Yuqian Jiang*, **Xiaohan Zhang**, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. LLM+P: Empowering Large Language Models with Optimal Planning Proficiency. *ArXiv, 2023 (*Equal Contribution)*
3. Yan Ding, Cheng Cui, **Xiaohan Zhang**, and Shiqi Zhang. GLAD: Grounded Layered Autonomous Driving for Complex Service Tasks. *ArXiv, 2022*

JOURNAL ARTICLES

1. **Xiaohan Zhang**, Saeid Amiri, Jivko Sinapov, Jesse Thomason, Peter Stone, and Shiqi Zhang. Multimodal Embodied Attribute Learning by Robots for Object-Centric Action Policies. *Autonomous Robots, 2023*
2. Yan Ding, **Xiaohan Zhang**, Saeid Amiri, Nieqing Cao, Hao Yang, Chad Esselink, and Shiqi Zhang. Integrating Action Knowledge and LLMs for Task Planning and Situation Handling in Open Worlds. *Autonomous Robots, 2023*
3. Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang. Learning to Ground Objects for Robot Task and Motion Planning. *IEEE Robotics and Automation Letters (RA-L), 2022*

CONFERENCE & WORKSHOP PAPERS

1. **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuqian Jiang, Yuke Zhu, Peter Stone, and Shiqi Zhang. Symbolic State Space Optimization for Long Horizon Mobile Manipulation Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023*
2. Yan Ding*, **Xiaohan Zhang***, Chris Paxton, and Shiqi Zhang. Task and Motion Planning with Large Language Models for Object Rearrangement. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023 (*Equal Contribution)*
3. **Xiaohan Zhang**, Yan Ding, Saeid Amiri, Hao Yang, Andy Kaminski, Chad Esselink, and Shiqi Zhang. Grounding Classical Task Planners via Vision-Language Models. *ICRA Workshop on Robot Execution Failures and Failure Management Strategies, 2023*
4. Thomas Lew, Sumeet Singh, Mario Prats, Jeffrey Bingham, Jonathan Weisz, Benjie Holson, **Xiaohan Zhang**, Vikas Sindhwani, Yao Lu, Fei Xia, Peng Xu, Tingnan Zhang, Jie Tan, and Montserrat Gonzalez. Robotic Table Wiping via Reinforcement Learning and Whole-body Trajectory Optimization. *IEEE International Conference on Robotics and Automation (ICRA), 2023*
5. **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuke Zhu, Peter Stone, and Shiqi Zhang. Visually Grounded Task and Motion Planning for Mobile Manipulation. *IEEE International Conference on Robotics and Automation (ICRA), 2022*
6. **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang. Planning Multimodal Exploratory Actions for Online Robot Attribute Learning. *Robotics: Science and Systems (RSS), 2021*
7. Kishan Chandan, Jack Albertson, **Xiaohan Zhang**, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang. Learning to Guide Human Attention on Mobile Telepresence Robots with 360 Vision. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021*
8. Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang. Task-Motion Planning for Safe and Efficient Urban Driving. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020*

ACADEMIC SERVICES

Program Committee Member:
- AAAI Conference on Artificial Intelligence (AAAI), 2023, 2024

Journal Reviewer:
- IEEE Robotics and Automation Letters (RA-L), 2022, 2023

Conference Reviewer:
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021, 2022, 2023